

# **CROWN AND BRIDGE**

## **FULL-METAL WITH FACING CROWN PREPARATION**

**Lecture six**

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# Full metal crown with facing (metal-ceramic crowns)



## **It is a full (complete) crown having:**

- **The labial or buccal surface (or all surfaces) covered by a tooth colored material (acrylic or porcelain). It consists of a ceramic layer bonded to a thin cast metal coping**
- **It combines the strength of full metal and the cosmetic effect of the tooth colored material.**
- **It can be used on the anterior and posterior teeth.**
- **It is not a conservative type of crown because it requires excessive tooth reduction to provide enough space for the metal and the facing material**





## Indications:-

1-Improvement of esthetic (carious teeth, malposed teeth, peg shaped lateral incisor, colored teeth).



# **Indications:-**

**2-Fractured teeth.**

**3-Teeth with large fillings.**



# Indications:-

4-AS A BRIDGE RETAINER ESPECIALLY IN LONG SPAN BRIDGE.





**5- TO PROTECT ENDODONTICALLY TREATED TEETH  
WITH SUFFICIENT TOOTH STRUCTURE  
(STRENGTHENING WEAK TOOTH STRUCTURE).**





## **Contra-indications:-**

- 1-Teeth with large pulp in young people (because of possibility of pulp exposure during preparation).**
- 2-Teeth with short crowns due to more occlusal preparation required.**
- 3-Patient with bad oral hygiene.**
- 4- Cannot accommodate a rest for removable partial denture because ceramic material may be fractured.**
- 5-Intact buccal wall where a more conservative retainer like three quarter can be used.**



# DISADVANTAGES OF PFM CROWN

1. Removal of substantial (important) amount of tooth structure.
2. Subject to fracture because of the brittle nature of porcelain.
3. Shade selection can be difficult due to metal layer below.
4. Inferior esthetic compared to porcelain jacket crown.
5. Discoloration of the gingival margin may occur with time.
6. More expensive.







## **Steps of preparation on posterior teeth:**

**The same principles and steps used for full metal crown preparation should be followed except that:**

**1- The buccal preparation should be deeper (more reduction) than that used for full metal crown preparation, to provide enough space for the metal and ceramic facing material to have proper shape and shade of the ceramic facing .**

**2- The finishing line should be beveled shoulder on the buccal surface and chamfer on the other surfaces.**

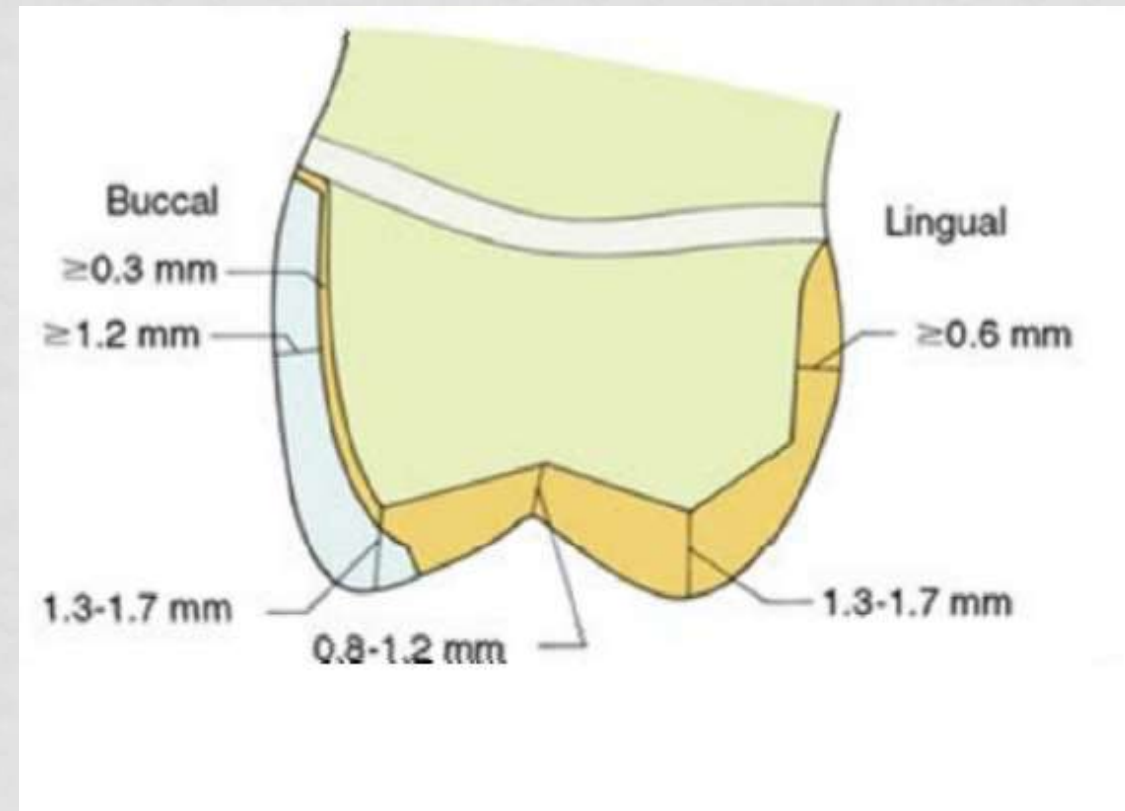
**Due to less esthetic requirements in posterior cases supragingival finishing line may be used.**

**3- While in anterior cases and retention requirements, subgingival finishing lines may be used in anterior cases.**

**Deep facial reduction to provide enough space for the metal coping and porcelain and shallower reduction on the other surfaces covered with metal only.**

**Bevel-shoulder and bevel-radial shoulder, or heavy chamfer can be used as a gingivofacial finishing lines, whereas chamfer or knife edge finishing lines is used for the remaining surfaces covered with metal only.**

**Since this restoration is a combination of metal and porcelain, tooth preparation likewise is a combination.**



**Lingual preparation done by the same procedure described for full-metal crown (using round-end fissure bur)**

**As the lingual chamfer is developed, extend it buccally into the proximal to join with the interproximal shoulder placed earlier, a lingual chamfer is prepared to allow adequate space for metal.**

**The margin should follow the soft tissue contour.**

**The lingual aspect of the proximal axial walls, as well as the lingual surface can also be reduced with the **torpedo diamond bur**.**





**Posterior metal  
with  
buccal facing  
crown**





# Preparation for anterior teeth

## Armamentarium:

The instruments needed to prepare teeth for a metal –ceramic crown include:

- Depth orientation groove bur
- **Round-tipped bur:** regular grit for bulk reduction and fine grit for finishing) .
- Football –or- wheel –shaped diamond (For lingual reduction of anterior teeth).
- Flat –ended, tapered diamond (for shoulder preparation).
- Finishing stones.



# FABRICATION OF SILICONE INDEX

The silicone index acts as a guide to check the amount of tooth structure removal.



## 1-Incisal reduction:

**(A) Place three depth grooves (1.8mm deep) on the incisal edge of an anterior tooth. This will provide the needed reduction of 2mm and allow finishing.**

**(B) The diamond fissure should be perpendicular to the long axis of the tooth.**

**(C) Remove the islands of remaining tooth structure between the depth grooves.**

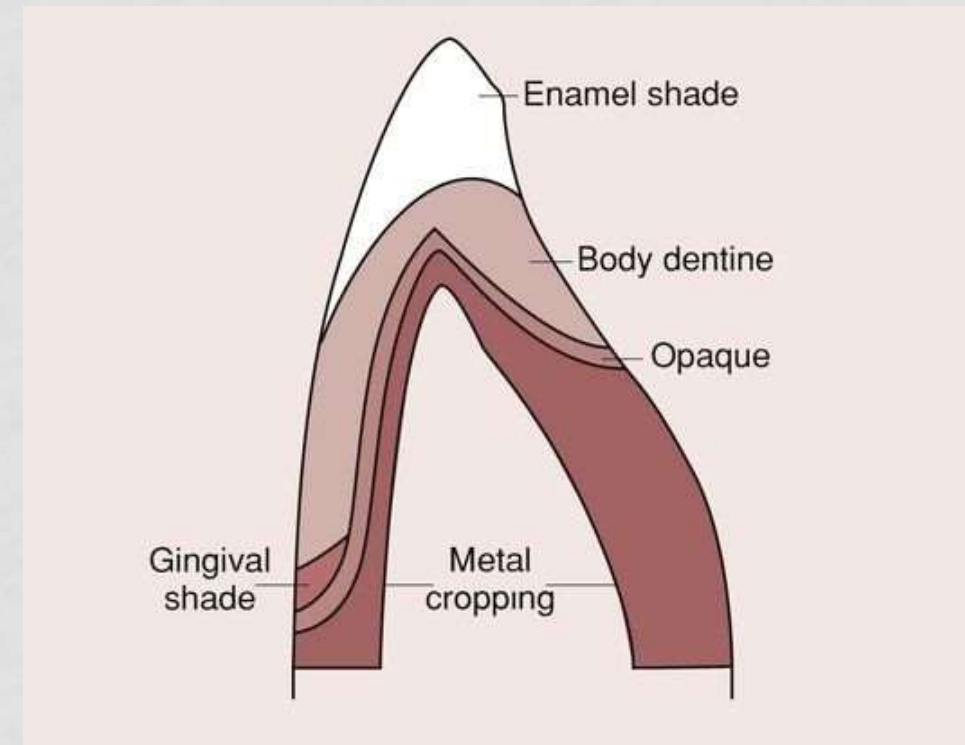
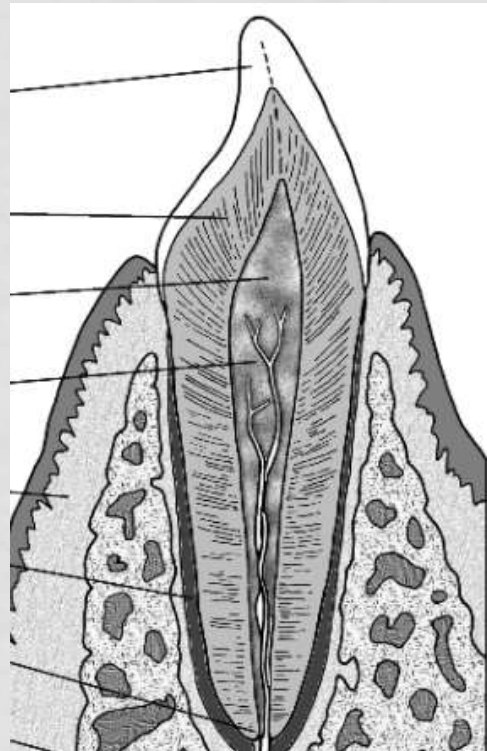
**(D) Flat-end tapered diamond bur is used, placed parallel to the incisal inclination (with a slight palatal inclination in the upper incisors and labial inclination in the lower incisors).**





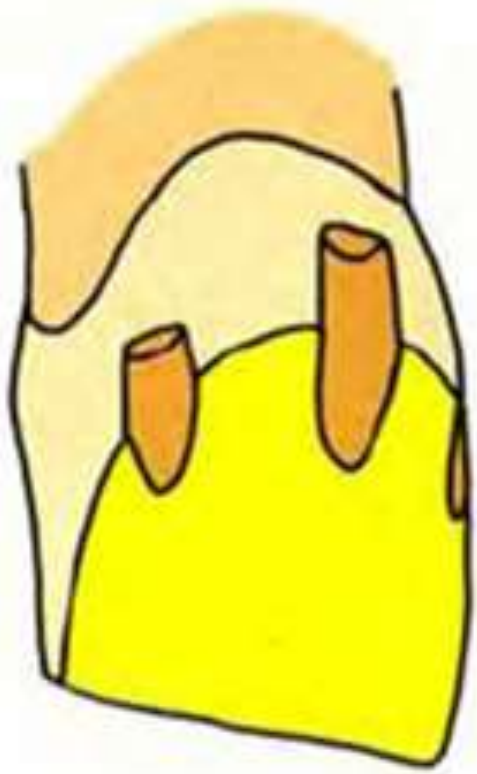
The completed reduction of the incisal edge should allow **2mm** for adequate ceramic material thickness to permit building incisal translucency in the completed restoration .

Caution must be taken to **avoid excessive incisal reduction** because shortening the axial walls lead to inadequate retention and resistance because most of the retention in anterior teeth is derived from the proximal walls.





**Incisal reduction should be done first to shorten the crown make it easy for the reduction of the other axial surfaces to be prepared to the finishing line (crown length = bur length) .**



# LABIAL REDUCTION

**PFM crown preparation requires deep facial reduction to give enough space for metal and porcelain, and thus avoiding over contouring and poor esthetic which would expected to occur when no enough tooth structure is removed.**

**The amount of labial reduction is  
1.5-2 mm.**

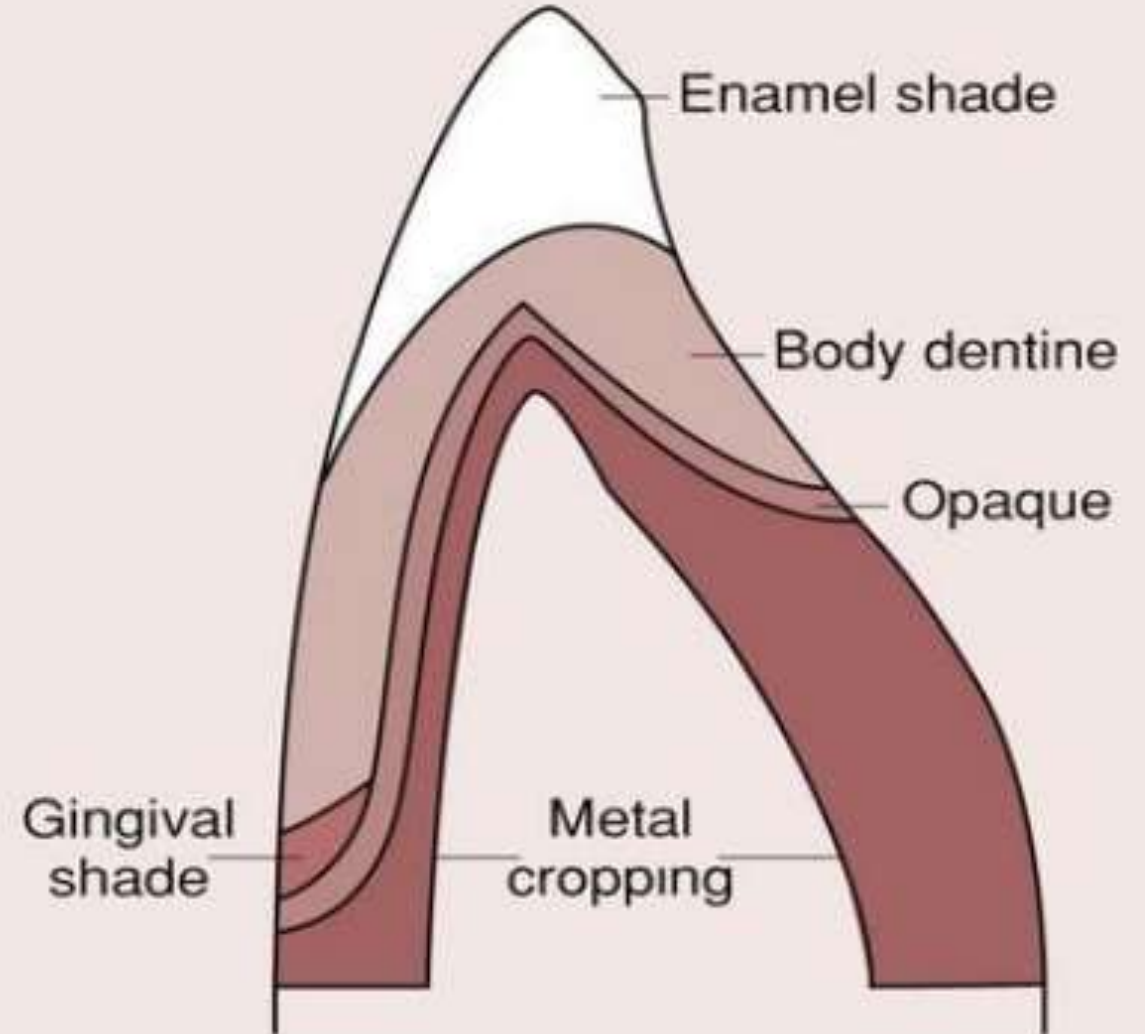


## **ADVANTAGES OF ADEQUATE REDUCTION (DEEP FACIAL REDUCTION):**

**1. The restoration will properly contour (effect on esthetic and gingival health).**

**2. The shade & translucency of the restoration will match that of the adjacent natural tooth.**

- 0.5 mm for the metal coping.**
- 1 mm for porcelain (0.2 mm opaque layer, 0.5 mm body “dentin” layer, and 0.3 mm incisal “enamel” layer).**



Because of the anatomy of the tooth labially, it should be reduced in **two planes** corresponding to the two geometric planes of the labial surface: a gingival plane and an incisal plane.

Advantages of two plane (step) reduction

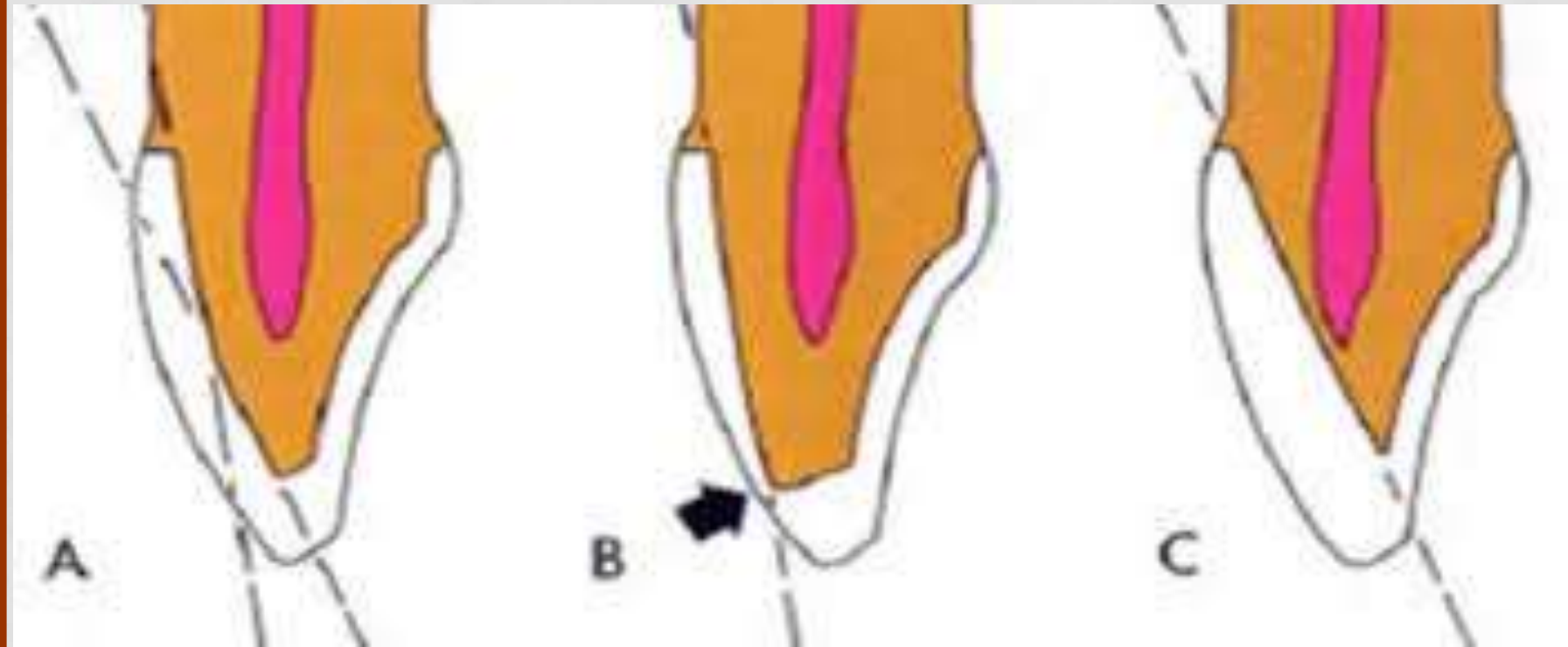
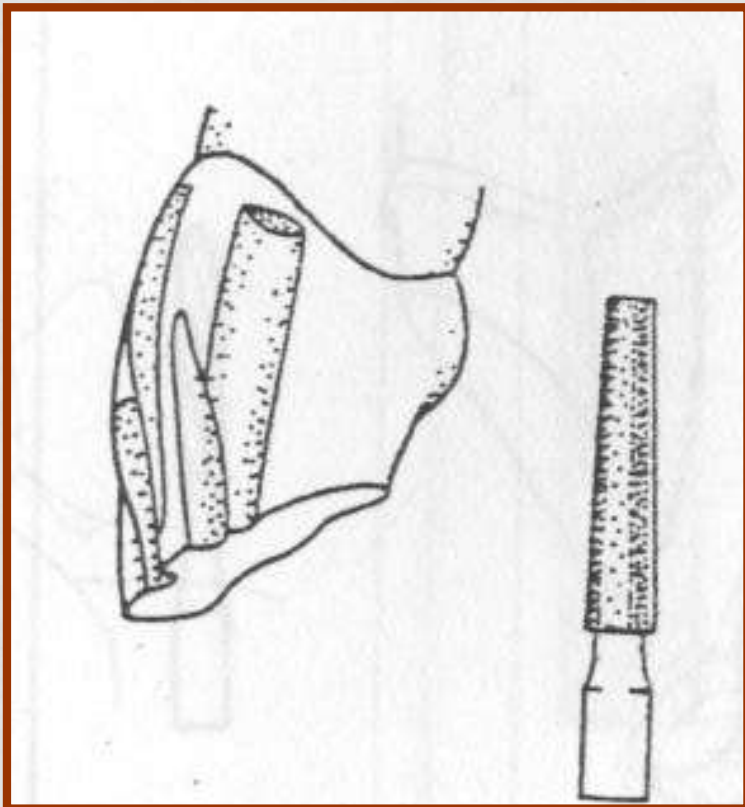
1. To follow the anatomy of the surface.
2. To avoid hitting the pulp.
3. To give enough space for the metal and porcelain layers, so that avoiding poor esthetic or over contour.



## **2-Labial reduction.**

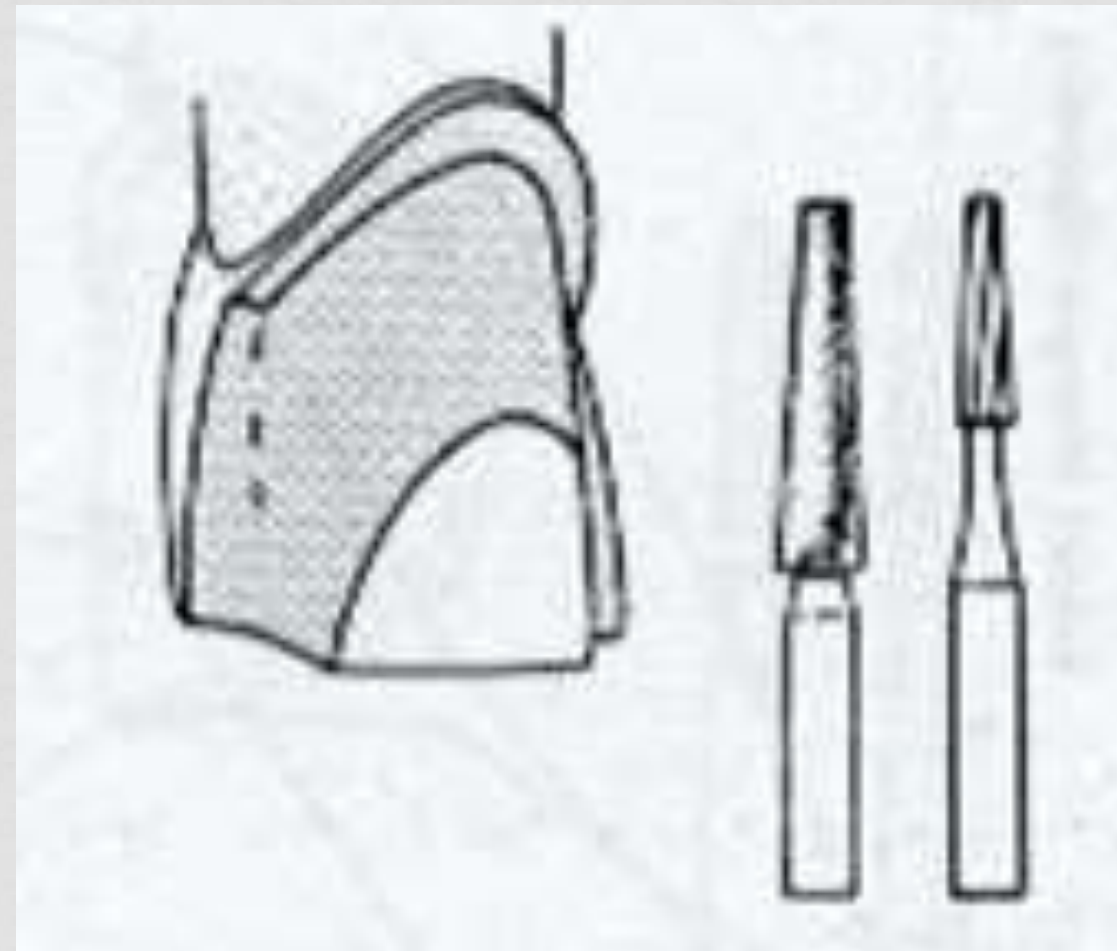
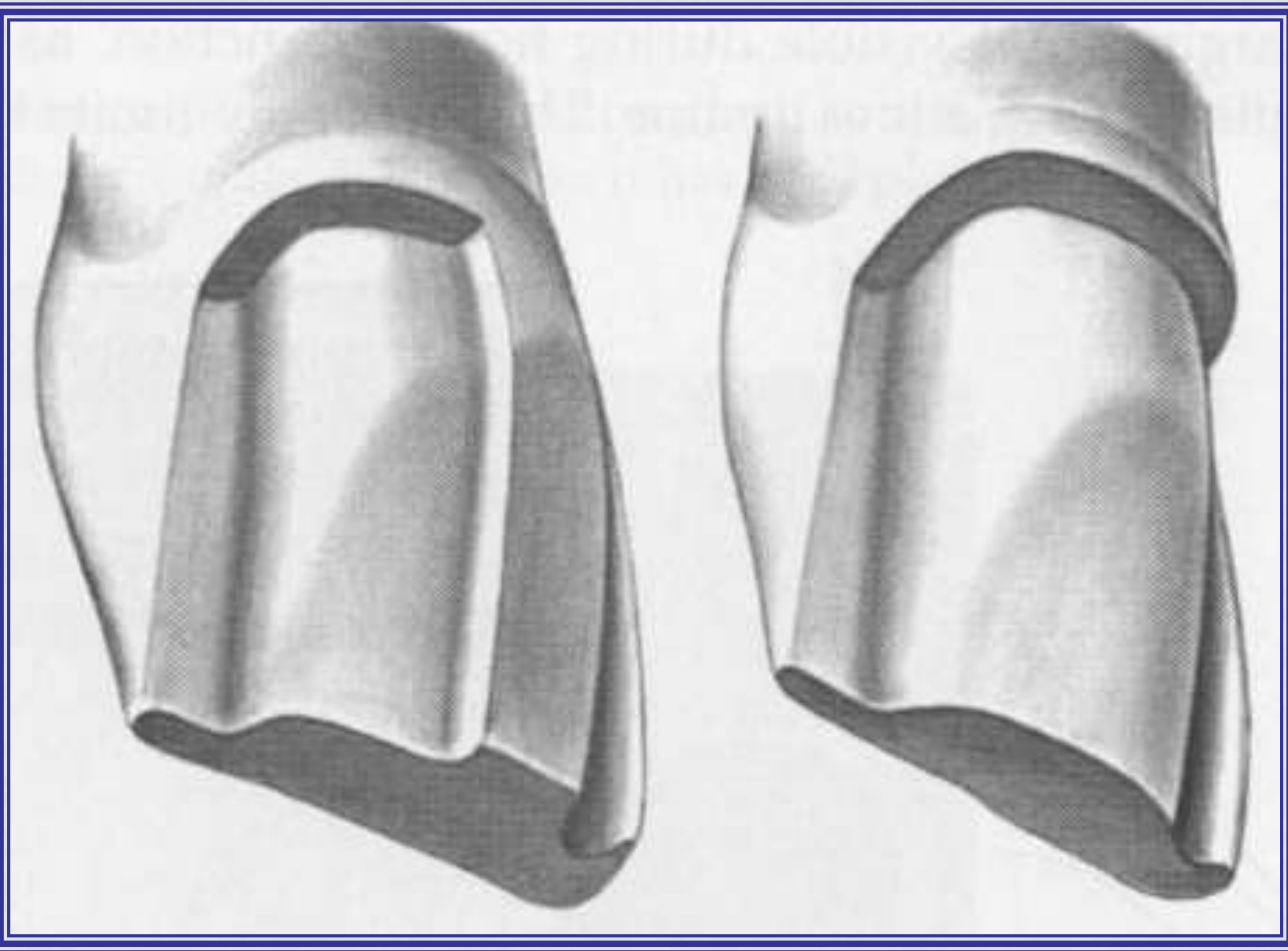
### **A) Cervical (Gingival) portion**

Depth grooves of 1 mm. depth are placed parallel to the long axis of the tooth, and by moving the bur with the inclination, this portion will be prepared. This will determine the path of insertion of crown restoration.



## B) Incisal portion

Depth grooves should be placed parallel to the inclination of this area, or follow the normal facial contour. This will provide the space needed for the proper thickness of porcelain veneer. By this way of preparation pulp exposure will be avoided.



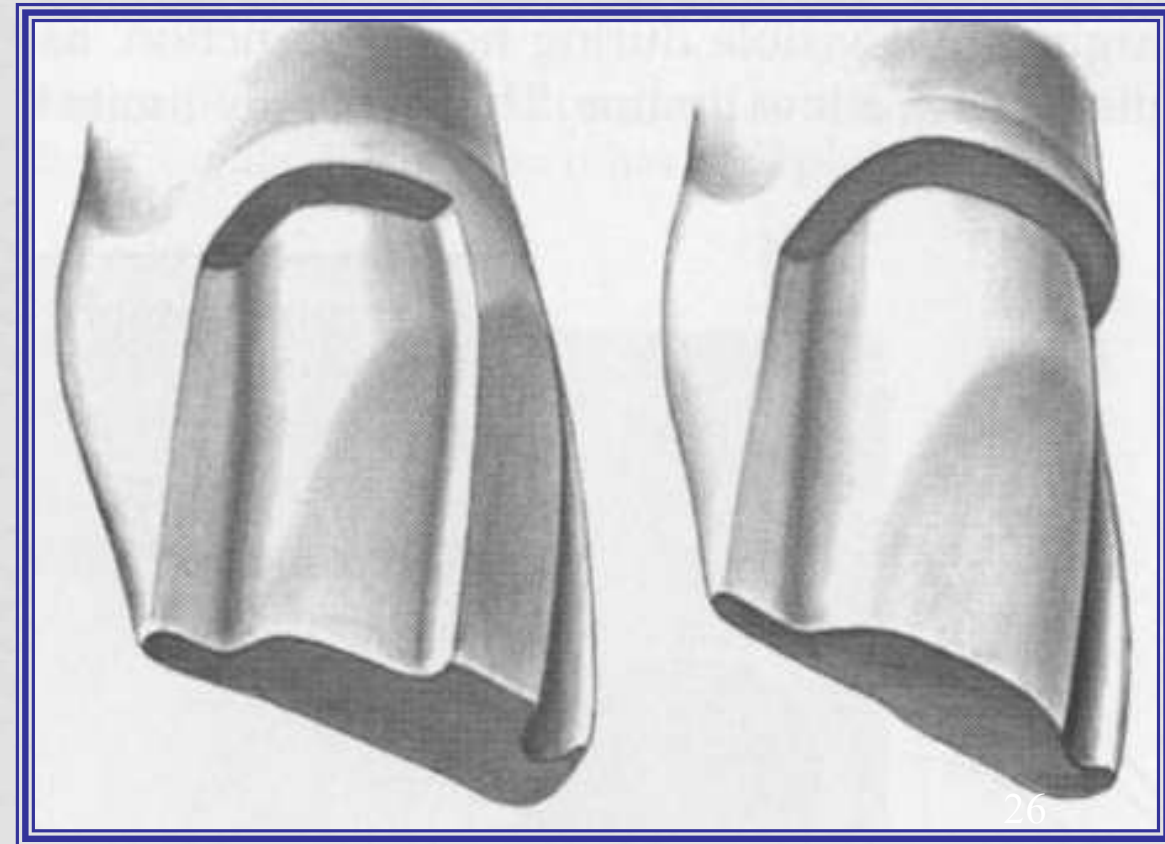
## **Incisal portion**

- \* The finishing line should be prepared as beveled shoulder so that the bevel will extend under the gingiva to hide the metal portion of the crown margin.
- \* The facial shoulder should extend at least 1mm lingually into the proximal contact, to ensure esthetics (hiding metal portion).

## **The resulting wing will provide:**

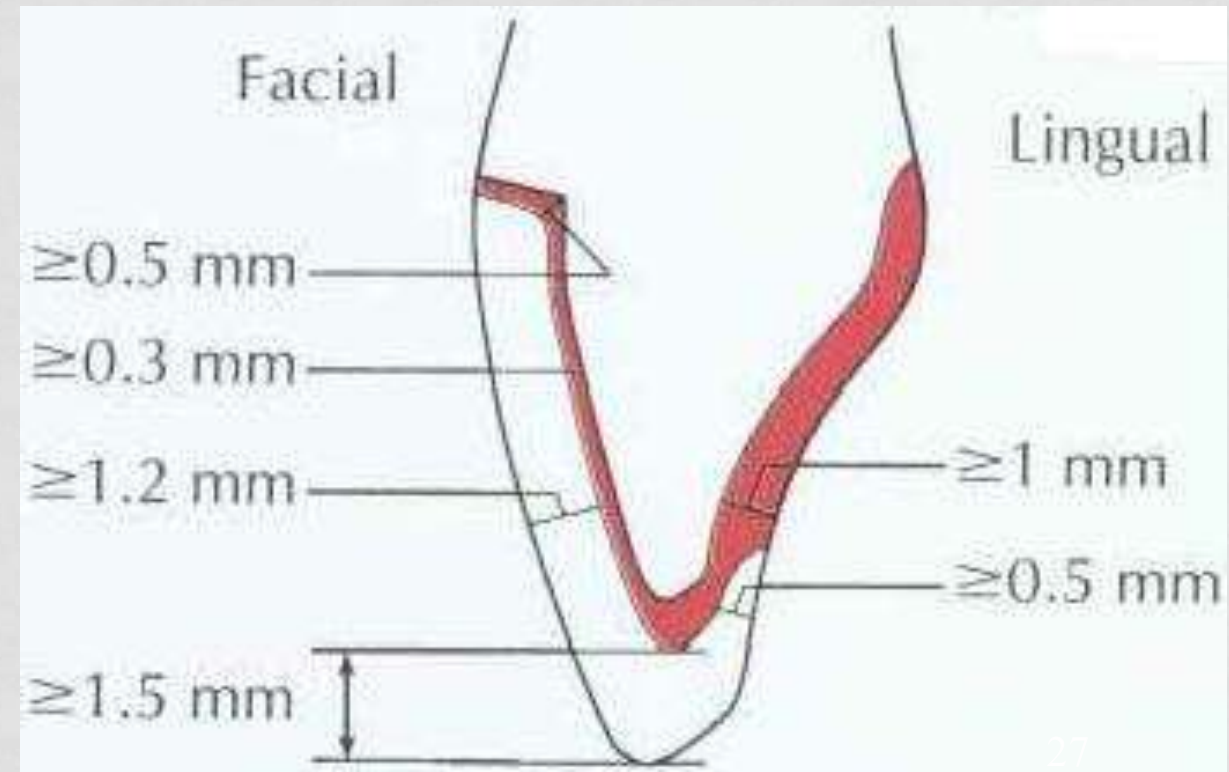
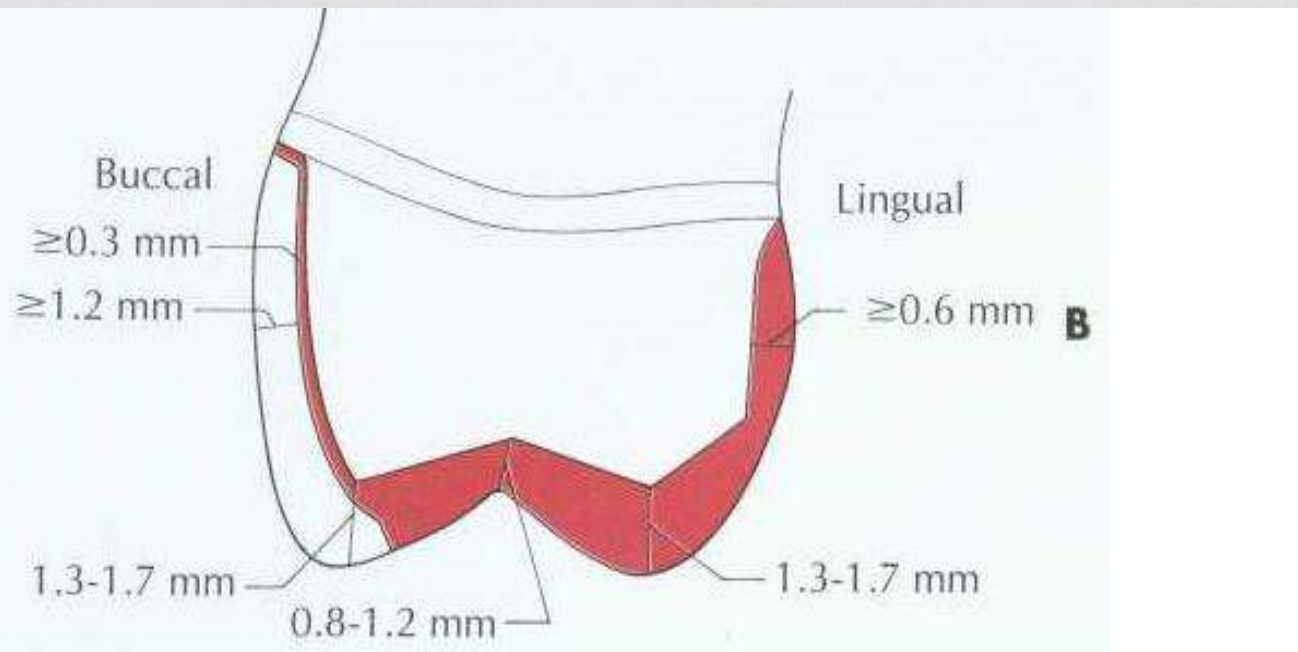
- 1- Some resistance to rotation.
- 2- Conserve tooth structure.
- 3- Retention.

It is important that the portion of each wing that faces labially has the same Inclination of the gingival portion of the labial surface to avoid creating difficulty in crown seating.



**The reduction of the facial surface should produce sufficient space to accommodate the metal substructure and porcelain veneer.**

**A minimum of 1.2 mm is necessary to make it easy for the technician to produce a restoration with satisfactory appearance (1.5 mm is preferable).**





### **3-Lingual reduction.**

The lingual preparation is divided to:-

#### **A) Cingulum portion.**

Due to metal coverage of this surface 0.5 mm. depth groove is placed parallel to the long axis of the tooth in the center of the cingulum,

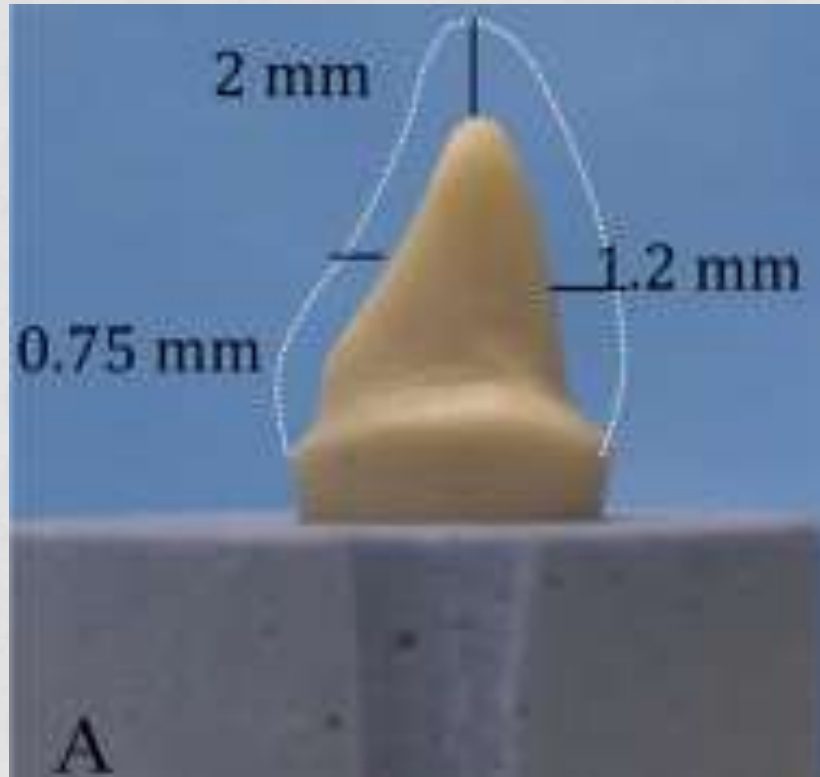
and by moving the bur mesially and distally the area will be reduced producing smooth chamfer finishing line .



### **b) Lingual fossa portion.**

The remaining lingual surface should be reduced using a wheel diamond bur, the junction between the lingual fossa and the cingulum portion should be preserved to increase retention and resistance by increasing the surface area.

Finally the line angles should be smoothed and rounded to facilitate the steps of crown construction.



## 4- Proximal reduction.

Preparation of the proximal surfaces is done in the same manner as in the full metal crown preparation.

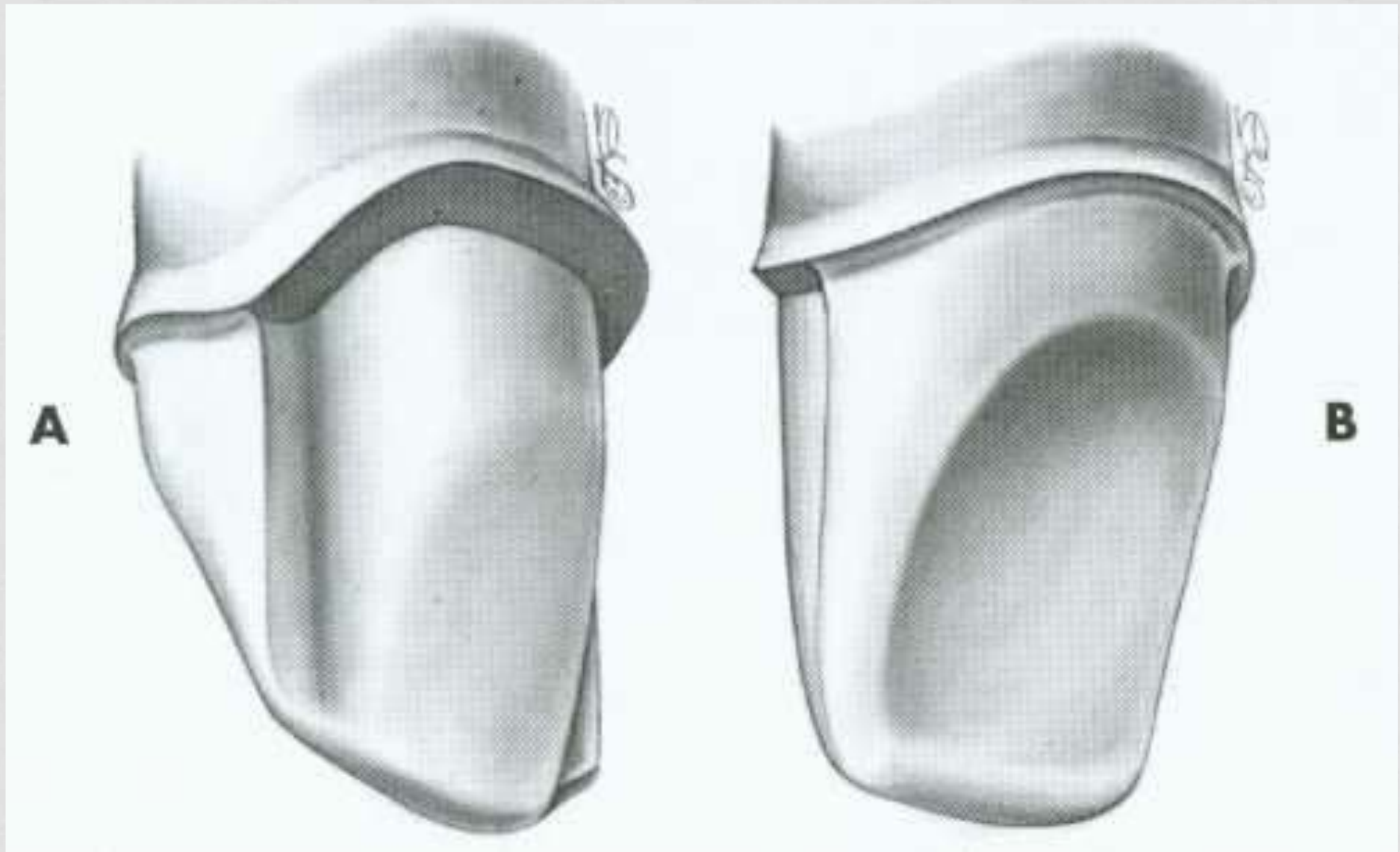
The finishing line for the proximal and lingual surfaces should be chamfer above the gingiva.



## **Evaluation:**

- \*Avoid creating an undercut between the facial and lingual walls.
- \*Excessive convergence should be avoided, because this may lead to pulp exposure.
- \*Avoid any unsupported tooth structure at the shoulder and chamfer finish lines.
- \*The completed chamfer should provide 0.5 mm of space for the restoration at the margin strength.
- \*The chamfer should be continuous with the interproximal shoulder.





**Complete preparation**

# Preparation on canines

**The same steps of preparation on incisors are followed for canines with only two exceptions which are:-**

**1-For the cusp area the depth groove should be placed at the tip of the cusp and by moving the bur mesially and distally along the slopes of the cusp this area will be reduced, therefore any horizontal straight reduction should be avoided at this area.**

**2-For the palatal or lingual surface, the final preparation should preserve the lingual anatomy of the lingual ridge and 2 lingual fossae.**



# **FULL METAL CROWN WITH ACRYLIC FACING**

It is a full metal crown whose labial or buccal surface is covered with tooth colored acrylic resin. It has been widely used previously before the use of porcelain as a facing material, but still used nowadays due to its lower cost as compared to PFM.

It combines the strength and accuracy of full metal crown with the esthetics of tooth-colored acrylic resin.

It is less expensive than PFM crown.

The preparation involves deep facial reduction to provide enough space for both metal and facing material.



## **FULL METAL CROWN WITH ACRYLIC FACING**

The finishing line is shoulder with bevel facially (labially or buccally) and chamfer or knife edge for the other surfaces. When esthetic is critical, subgingival positioning of the finish line is recommended.

The main disadvantages of this type of crown are related to the acrylic facing material, including discoloration with time, wearing, poor bonding with metal and poor compatibility of the acrylic resin with the gingival tissue .



Thank You

Any question